KisanNiti: Cooperative Post-Harvest Selling for Farmer Producer Organizations with Mandi Price Forecasting, Storage Sensing and Fair Allocation

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Abstract. **DRAFT FOR AUTHOR REWRITE.** Drafted by Claude from verified results only. The sentence on the basmati forecast comparison is a marker: it is replaced by the result of the single pre-registered test-season run, whatever that result is. The authors rewrite the abstract in their own words.

A Farmer Producer Organization (FPO) has to decide where, when and how much of its members' produce is sold, and the tools that exist treat forecasting, storage and logistics separately. KisanNiti connects them in five stages that an FPO manager follows: collecting and cleaning prices and farmer records, proposing a selling plan, reviewing it, telling each farmer in their own language, and recording what was sold. A regime test separates price-bound from market crops. For Paddy (Common) in Punjab, at least 96.1% of the reported prices in every season from 2020 to 2025 equal the minimum support price, and a rule that predicts the support price had the lowest error (Rs 1.26 per quintal); no learned model beat it. For basmati, whose prices move, a forecast comparison of eight methods was fixed in a pre-registered protocol before any test result existed. **[Result of the held-out season to be inserted.]** An exact fair-allocation model shares each mandi's daily limit among members: in 60 synthetic plans at real prices, every plan was valid and the smallest share a farmer received rose from 0.0% under the nearest-mandi rule to 68.6%, at a cost of Rs 5.70 per quintal. The system was built and tested end to end (1,372 automated tests, a 14-step run on fictional data); no real farmer data or user study is included.

Keywords: Farmer Producer Organizations · Mandi price forecasting · Minimum support price · Fair allocation · Integer programming · Internet of Things

1 Introduction

DRAFT FOR AUTHOR REWRITE. Drafted by Claude from the v2 text. One author rewrites this section in their own words and adds what only the team knows: why we started with Patiala, what FPO staff told us, and what went wrong in v2. Delete this box when done.

A Farmer Producer Organization (FPO) is a registered body of farmers that can pool, market and sell its members' produce [42]. Selling the produce after harvest is where many FPOs struggle. For each member lot, the FPO has to choose a mandi, a day and a

quantity, and these choices depend on each other. The mandi with the best price may be the farthest from the village. A later selling date may pay more, but grain kept in storage can heat up or gain moisture. When many members deliver in the same week, a single mandi can receive more than it can weigh and unload in a day.

Losses after harvest are large in India. A 2024 ICRIER policy brief puts them at about Rs 1.53 trillion of food a year, using a national NABCONS survey from 2020–2022, and it names storage as one of the causes [23]. [Literature session: re-check this figure against the brief before submission.]

Two public sources give an FPO most of the price information it needs. AGMARKNET publishes the daily arrivals and the minimum, maximum and modal prices reported by each mandi [17]. For paddy and wheat, the government also announces a minimum support price (MSP) before each marketing season [16,15]. For Paddy (Common) in Punjab, the mandi prices hardly differ from it. Across the 20 mandis we studied, at least 96.1% of the reported prices in every season from 2020 to 2025 were exactly the Grade-A MSP. Basmati paddy has no MSP and is sold at market prices that differ by variety, so for basmati the choice of mandi and the day of sale can change what a farmer earns. It is the crop we use to test price forecasting and the hold-or-sell advice. Wheat is also bought at the MSP during procurement. After that window, Punjab mandis report very few wheat prices, too few to forecast held wheat from mandi data, so held wheat is left out of this study (Sect. 5). [Cite a source for basmati pricing. Give the off-season wheat row counts from the L1 download manifests as \kn{} macros, not typed by hand (decision CHAT-32).]

KisanNiti is the decision-support system we are building for FPO staff in the Patiala area. It reads AGMARKNET prices, the FPO's records of member lots and readings from a grain-storage sensor, and proposes a selling plan that a manager reviews before farmers are told on WhatsApp (Fig. 1). The paper makes five contributions. A regime test, the share of reports equal to the support price, tells price-bound from market crops and is applied to Paddy (Common) and basmati (Sect. 4.1). For basmati, a forecast comparison of eight methods with prediction ranges was fixed in a pre-registered protocol and is evaluated once on a held-out season (Sect. 4.2). An exact allocation model shares mandi limits fairly among members and is compared with five simpler selling rules (Sect. 4.3). A low-cost storage-sensing rule is tested on simulated scenarios (Sect. 4.4). Finally, the five stages from data to recorded sales are built and tested as one system (Sect. 4.5). The claims are limited to what each experiment supports; Sect. 5 lists what is not shown.

Section 2 reviews related work. Section 3 describes the system and our methods, Sect. 4 reports the experiments, and Sect. 5 states the limits of this study and the next steps.

2 Related Work

Price forecasting with Indian market data. Most Indian studies take daily or monthly prices from AGMARKNET and compare one family of models against ARIMA. Paul et al. forecast brinjal prices in 17 Odisha markets and found a generalised regression neural network more accurate than ARIMA in most markets [40]. Manogna and Mishra worked with exchange spot prices of five oilseed and guar commodities and reported that

an LSTM tracked the direction of price changes better than a time-delay network [34]. The largest comparison we found covers 23 commodities in 165 markets from 2010 to 2024, where LSTM and GRU models had the lowest errors [33]. Nayak et al. added rainfall and temperature to tomato, onion and potato series and found that deep models with these inputs beat statistical and tree-based ones [36]. Two recent studies are closer to our crops. Bhoi et al. fitted six time-series models to monthly basmati prices from 74 Punjab markets and reported forecasts as probability bands rather than single values [9]. Balantrapu et al. studied all-India monthly wheat and maize prices and asked what it means for a series to be anchored to the minimum support price (MSP). For wheat, a naive forecast that repeats the last price was almost as good as their best model [6]. Our v2 paddy result points the same way.

Graph and pretrained models. Graph models forecast many related series at once. Özden and Bulut used a spectral temporal graph network on 1,791 days of Turkish vegetable prices and trade volumes [39], and Min et al. found that StemGNN and T-GCN beat a stacked LSTM on four South Korean crops when prices and weather were modelled together [35]. T-GCN and its attention variant A3T-GCN were built for road traffic [47,5]; we use A3T-GCN with mandis as nodes. Pretrained "foundation" forecasters are newer. Chronos is trained on a large mix of public series and gives probabilistic forecasts without task-specific training [3]. A 2026 preprint reports that such models beat classical and deep baselines, and USDA's own forecasts, on monthly US crop prices [45]. We found no test of these models on AGMARKNET data.

Selling time and storage. Price support shapes how Indian farmers sell. In the 70th round of the national sample survey, only 23.7% of farmers knew the MSP of their kharif crop and 20.0% that of their rabi crop [1]. Small farmers were less likely to sell to procurement agencies [14], and a later count put the share of sellers reached by MSP at 13% for paddy and 16% for wheat [24]. Field studies in Africa explain why farmers sell at harvest. Timely credit let Kenyan farmers store maize and sell later at a 29% return [10]. In Benin, households that stored to sell later stored less when they expected higher storage losses [28]. Across 30 African countries the lean-season price failed to rise above the harvest price in 15–25% of years, so holding grain carries real price risk [11]. Tripathi et al. built the only decision tool we found that weighs storage, rent and physical loss against a price forecast; it advises individual producers and was simulated on tomato, onion and potato markets in north India [43]. Market choice matters as well: Karnataka's unified market platform raised the modal price of paddy by 5.1% [30].

Storage losses and sensing. Farm storage losses in North India are small for the first few months. In a 2022 survey by ICRIER, measured farm storage loss was 0.39–0.50% for wheat in Madhya Pradesh and zero for paddy in Punjab, where farmers sell soon after harvest; the authors expect losses to rise after about three months [22]. An older Karnataka survey found the largest farm-level losses of rice and wheat at the storage stage [7]. Low-cost storage monitors log temperature and humidity and raise alarms. Islam et al. watched hermetic paddy bags remotely for 125 days [27], and Indian prototypes send SMS alerts from warehouses or switch fans at set thresholds [32,26]. A recent review

Table 1. Features reported by representative studies and implemented in KisanNiti. D: Indian mandi data; M: MSP check; R: forecast ranges; H: hold-or-sell with storage cost; S: storage sensing used in the decision; C: shared mandi capacity; F: fairness criterion; A: manager approval; W: farmer messages. Y: reported; –: not reported in what was read. The KisanNiti row lists what is built and tested at the time of writing (ranges come from baseline forecasts; hold-or-sell is not built).

Study	D	M	R	H	S	C	F	A	W
Manogna et al. [33]	Y	–	–	–	–	–	–	–	–
Bhoi et al. [9]	Y	–	Y	–	–	–	–	–	–
Balantrapu et al. [6]	Y	Y	–	–	–	–	–	–	–
Min et al. [35]	–	–	–	–	–	–	–	–	–
Tripathi et al. [43]	–	–	–	Y	–	–	–	–	–
Islam et al. [27]	–	–	–	–	–	–	–	–	–
Anoop et al. [2]	–	–	–	–	–	Y	–	–	–
Kurokawa et al. [29]	–	–	–	–	–	–	Y	–	–
Fafchamps and Minten [19]	–	–	–	–	–	–	–	–	Y
KisanNiti v2 (tested)	Y	Y	–	–	–	Y	–	–	–
KisanNiti (this paper)	Y	Y	Y	–	Y	Y	Y	Y	Y

covers machine learning and IoT for stored grain [4]. None of these systems passes its readings to a selling decision.

Allocation and fairness. Mixed-integer models already plan agricultural logistics. Urra-Calfuñir et al. assigned olive harvests to fixed and mobile mills for small growers in Chile [44]. Gámez-Albán et al. routed a cooperative's first-mile pick-ups in Greece [20]. Anoop et al. allocated food-grain rail freight in India with penalties on rakes, capacity use and weekly limits [2]. These models minimise cost or maximise total profit and do not say how gains are shared between growers. Operations research offers tools for that. Bertsimas et al. bound the efficiency lost under max-min fairness [8], Ogryczak et al. survey lexicographic max-min (leximin) methods [38], Chen and Hooker show how to write these criteria in a mixed-integer program [12], and Kurokawa et al. used a leximin mechanism to allocate school classrooms with real district data [29]. We did not find a fairness criterion in any agricultural allocation paper we read.

FPOs and advice to farmers. India had 7,374 registered producer companies by March 2019, and the median one had only Rs 1.06 lakh of paid-up capital [37], so tools for FPOs must be cheap to run. Phone advice has a mixed record. SMS price and weather messages to farmers in 100 Maharashtra villages produced no significant change in prices received [19]. A voice advisory service for Gujarat cotton farmers changed input use but showed no clear gain in yield or profit [13].

Table 1 lists only what each paper reports. In the 38 studies reviewed, price forecasting, storage monitoring and allocation were studied separately. No study fed a forecast into a plan that shares mandi capacity among the members of an FPO, and none combined such a plan with a fairness rule or a manager's approval before farmers are told. KisanNiti addresses this combination for Punjab, where basmati is the main crop sold at market prices and nearly all reported prices of common paddy equal the MSP.

3 System Design and Methods

DRAFT FOR AUTHOR REWRITE. Drafted by Claude from the built and tested system and the pre-registered protocol (`layers/L3a_forecasting/PREREGISTERED_PROTOCOL.md`). The authors rewrite it in their own words, redraw Fig. 1 to match the five stages below, and check each statement against the code before submission.

KisanNiti is organised as the sequence in which an FPO manager uses it (Fig. 1): (1) farmer records, mandi prices and storage readings are collected and cleaned; (2) a forecast, a storage assessment and a fair allocation are turned into a proposed selling plan; (3) the manager reviews, edits and approves or rejects it; (4) each farmer is told in their own language; and (5) replies and actual sales are recorded and compared with the plan. Five rules hold throughout. Raw facts, derived values, system proposals, human decisions and what actually happened are stored separately and never overwrite each other. Quantities and money are exact (grams and paise, never floating point). Every FPO sees only its own data, enforced twice, in the query and in the database. Every change is an append-only record with the person and request that made it. Every failure is explicit: a missing contact, consent or language blocks a message with a stated reason instead of being guessed.

3.1 Stage 1: Data and Cleaning

Prices come from the AGMARKNET report *Date Wise Prices for Specified Commodity*, downloaded by hand for each month because the portal asks for a CAPTCHA. 29 months of Paddy (Basmati) for Punjab are filed with a manifest of checksums and row counts. A parser keeps every line with its source file and position, so that each row of the cleaned table can be traced to the line it came from. The 29 monthly files hold 6,678 rows (6,098 with a date and 580 without), and every row is accounted for. A mechanical check flags rows that look wrong (for example a minimum price above the maximum) but removes none. Where the portal shows a variety under the date of the row above it, the date is inferred; those rows are flagged and are the subject of the hand check below. Prices of one variety are summarised into a daily index: the median modal price over the mandis that reported the variety on that day, at least three mandis, no filling of missing days. The index is a market summary, not the price a farmer received. Farmer records (members, lots with exact registered quantity, places, and an append-only stock ledger) are entered by FPO staff. Storage readings arrive from a sensor node over an authenticated channel, are validated and stored once, and are assessed hourly (Sect. 3.3).

The cleaned release is imported into the product with an exact row reconciliation and the status `PENDING_HAND_CHECK`. It becomes `VALIDATED` only after a hand check of a stratified sample of 56 rows against the portal (problem-flagged rows, blank-date rows, extreme prices, ordinary rows, prices equal to a support value, and the largest arrivals). That check, and an independent spot check against a second source, decide the blank-date rule. This paper reports the data as unchecked until the check has been done.

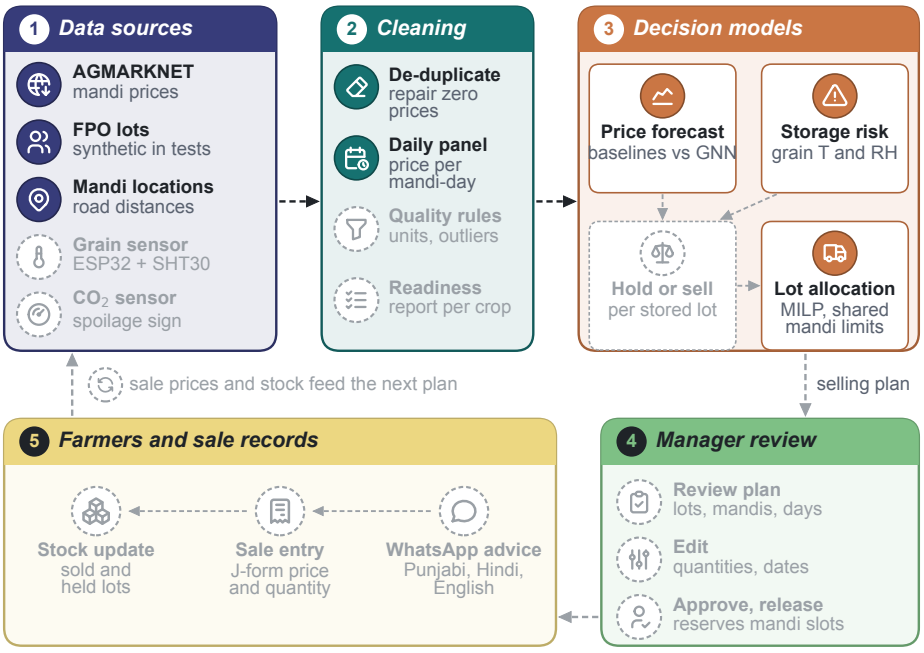


Fig. 1. KisanNiti architecture. Data sources (1) are cleaned into a daily mandi price panel (2). Decision models (3) turn prices, storage readings and lot records into a selling plan that the FPO manager reviews (4). Farmers receive the approved plan on WhatsApp, and their recorded sales update stock for the next plan (5). MILP: mixed-integer linear program; GNN: graph neural network; RH: relative humidity.

3.2 Stage 2: Forecasting, Storage Risk and Fair Allocation

Regime test. For each crop and season, the share of reported prices equal to the announced MSP is computed. A high share marks a price-bound crop, for which the announced MSP is used as the price and no forecaster is trained; a low share marks a market crop.

Pre-registered forecast comparison. For market-priced basmati (varieties 1121 and 1509), the daily index of Sect. 3.1 is forecast at horizons of 1, 7 and 14 calendar days (28 days secondary, 60 days descriptive only). Seasons 2021–22 to 2023–24 are used for the initial training, 2024–25 to tune and calibrate, and 2025–26 is held out and evaluated once; September and October 2026 form a separate prospective check. Each forecast uses only data up to its origin day, and an origin needs ten index days of its own season. Eight methods were fixed before any test result existed: the last value, a seasonal naive value, a five-day moving average, ETS, ARIMA, LightGBM, a small GRU and the pre-trained Chronos-2 model used zero-shot. Ranges of 80% and 95% come from analytic

intervals, quantile objectives, split-conformal residuals or the model's own quantiles. Accuracy is reported as the mean absolute error (MAE) in rupees per quintal and as MASE; each model is compared with the naive value by a Diebold–Mariano test with the Harvey–Leybourne–Newbold correction and a Newey–West variance, with Holm's correction over all 42 comparisons, plus a block-bootstrap interval of the relative change in MAE. A model is called better only if the adjusted $p < 0.05$, the interval excludes zero, and the gain exceeds the smallest gain the sample can detect; otherwise the result is "no detectable difference", and the product keeps the naive value. The protocol, its open choices and the evaluation code are in the project repository, and the code is tested on synthetic series with known answers (`layers/L3a_forecasting/`).

Fair allocation. A plan assigns each lot a mandi, a day and a quantity. Each mandi has a daily limit shared by all members. The solver is an exact mixed-integer program with a fixed hierarchy: allocate as much as possible, then maximise the smallest share any farmer receives (and so on, lexicographically), then give lots at risk of spoilage the earliest dates, then maximise the net return, then break ties deterministically. An independent checker re-validates every plan and a brute-force reference confirms the solver on small cases. Selling rules used for comparison are described in Sect. 4.3.

3.3 Storage Sensing and Risk Rule

DRAFT FOR AUTHOR REWRITE. Claude draft, 22 Oct 2026. Rewrite in the authors' words. The node in Fig. 2 is not assembled yet (kit not bought); everything marked built has been compiled and tested, and the rule has been run only on simulated readings (Sect. 4.4).

Basmati paddy is stored on the farm in jute bags until it is sold, often for months. To give the selling plan a signal about storage conditions, we place one probe inside a bag and read the air between the kernels. Figure 2 shows the path from that probe to the planner.

Node. The probe is a protected SHT30 humidity and temperature sensor. Its typical tolerance is ±2 %RH between 20 and 80 %RH, widening to ±3–4 %RH near the ends of the range, and ±0.2 °C from 0 to 65 °C [41]; the limits used below (65 and 70 %RH) lie in the ±2 %RH band. It is connected over I^2C (GPIO 21 and 22, 4.7 kΩ pull-ups) to an ESP32-WROOM-32 board [18] that stays outside the bag. Every 5 minutes the firmware takes one single-shot, high-repeatability reading and checks the CRC byte that the sensor sends with each value. A failed read is sent as an empty value with its error code, never as zero. Each message carries the device identifier, a random boot identifier and a sequence number; together these identify a reading uniquely, so a repeated delivery can be recognised. Readings taken before the clock is synchronised get their time later from the board's uptime, or stay without a time.

Transport and storage. The node sends JSON messages over the farm's phone hotspot to an MQTT broker on the FPO's laptop, using TLS with a separate password and topic permission for each device. A reading stays in a 288-sample buffer in the board's memory (24 h) until the server reports that it has stored it; a broker acknowledgement alone is

not enough. The server checks each message against a JSON schema, stores it once per (device, boot, sequence) and then sends the acknowledgement. The buffer is in RAM, so a power cut loses the readings that were not yet sent. The same firmware runs in the Wokwi simulator [46]; Wokwi has no SHT3x part, so we wrote a small custom chip that answers the sensor's I^2C commands with the same CRC, which lets anyone run the node without hardware. Readings from this twin are labelled synthetic.

Storage rule (STORAGE-DEMO-0.1). Grain exchanges moisture with the air between its kernels until the two are in equilibrium; that air's relative humidity divided by 100 is the grain's water activity a_w. The storage limits in the literature are stated on this scale: storage fungi need a relative humidity of at least 65 % and grow between about 10 and 40 °C [25], and grain kept in equilibrium with air below 65–70 % RH degrades slowly [31]. The rule therefore takes the median relative humidity and temperature of the last 24 h and assigns LOW for $a_w < 0.65$, MEDIUM for $0.65 \le a_w < 0.70$ and HIGH for $a_w \ge 0.70$. The level rises as soon as the median crosses a limit, but falls only after the median has stayed at least 2 points below the limit for 48 h, so that a reading near 65 % does not switch back and forth. Grain below 10 °C is moved one level down; grain at 25 °C or warmer with $a_w \ge 0.60$ is moved from LOW to MEDIUM. A separate note says whether the temperature favours insects ($\ge$ 25 °C), slows them (15–25) or stops most activity ($<$ 15); it does not change the level.

The rule returns UNKNOWN, with the reasons, when the probe is registered as hanging in shed air, during the first 6 h after the probe is pushed into a bag, when the newest usable reading is older than 15 minutes, or when fewer than 75 % of the expected readings of the last 24 h are usable. The same thresholds apply to basmati paddy and to wheat, because they are limits on water activity rather than on a crop's moisture content. The rule does not report moisture content, which would need sorption-isotherm constants that we could not verify, and it does not detect spoilage. It says whether the air in one bag favours mould and insects. The rule is part of the shared core package that the product also uses, and is covered by unit, boundary and property tests.

[Settling time (6 h) and the 2-point margin are desk-research assumptions. Replace them with the bench values from `firmware/BUILD_GUIDE.md` (H4) once measured, and cite the result.]

3.4 Stage 3: Manager Review and Approval

A proposed plan is stored unchanged. A manager may edit it; each edit becomes a new revision that is accepted only if the independent validator finds no violation, and the system's proposal is never rewritten. Approval requires the *approve* permission and an explicit acknowledgement of every warning, refers to one revision, and in one database transaction records the decision and reserves the exact quantities in the stock ledger. The reservation cannot exceed free stock, approval cannot be given twice, and two plans competing for one lot cannot both succeed; this was tested with parallel requests on PostgreSQL. A plan can also be rejected with a coded reason, and an unused reservation is released by a person.

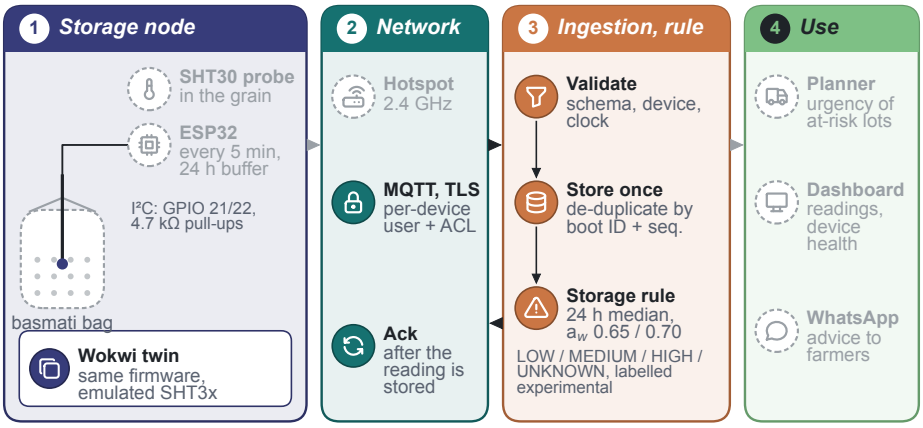


Fig. 2. Storage node and data path. An SHT30 probe in a bag of basmati paddy is read every 5 minutes by an ESP32 (1); readings travel over the hotspot by MQTT with TLS (2); the server validates them, stores each reading once and applies the storage rule STORAGE-DEMO-0.1 (3), whose category feeds the planner and the screens (4). The Wokwi twin runs the same firmware against an emulated sensor. Dashed grey parts are not yet built. I^2C: inter-integrated circuit bus; TLS: transport layer security; ACL: access-control list; a_w: water activity.

3.5 Stage 4: Telling the Farmers

After approval, one message per farmer is built from a versioned template in the farmer's recorded language (Punjabi, Hindi or English) and the approved facts only: farmer name, variety, quantity, mandi and date, with no price and no promise. A message can be prepared only if the farmer is active, has a phone number and has agreed to WhatsApp messages; otherwise it is blocked with the reason, and the checks are repeated at the moment of sending. A *preview* channel shows the exact text and a masked recipient without sending anything. For use without a messaging-platform account, the manager can instead press "Send to farmers", confirm with yes or no, and receive one link per farmer that opens the manager's own WhatsApp with the stored text already written; he sends it there and reports "sent" or "not sent" for each farmer. The system cannot see delivery, so that report is stored as the manager's statement and never as a delivery. Phone numbers are masked for every role that may not see them, and each number shown is logged.

3.6 Stage 5: Replies and Actual Sales

Staff enter farmers' replies as text; a reply is stored and shown but never read as acceptance. A sale is entered from the paper sale form (J-form) with lot, mandi, date, exact quantity and price; the amount is computed exactly with ties to even, a duplicate form number is refused, and a mistaken entry is voided by a compensating record. A sale on plan consumes its reserved stock; a sale off plan needs the person to choose which reservation to give up (a provisional rule, not yet validated with an FPO). The system then

Table 2. Paddy (Common) prices in the 20 selected Punjab mandis, September to December of each season. SD = standard deviation. Generated from the v2 result files.

Season	MSP Grade A (Rs/q)	Trading dates	Prices equal to MSP (%)	SD of prices (Rs/q)
2020	1,888	70	98.3	44.3
2021	1,960	55	99.1	80.5
2022	2,060	60	98.5	16.5
2023	2,203	70	96.1	45.5
2024	2,320	66	98.8	29.5
2025	2,389	61	99.0	21.6

reports planned against actual quantity, price and share per farmer, and keeps forecast outcomes of fictional and real sales apart.

4 Results and Discussion

4.1 Paddy (Common): Prices Stay at the MSP

DRAFT FOR AUTHOR REWRITE. Drafted by Claude from the v2 results (Sect. 4.1–4.2 of v2), rewritten to avoid the patterns the college checker flagged. The author who rewrites it should say what they noticed in the raw data (for example, which mandis left the MSP in 2021 and 2023, and why).

Table 2 describes the prices of the 20 mandis that reported in all six seasons. Across 382 trading dates, these mandis reported a price on 4,626 mandi-days (60.5% of the possible mandi-days). In the 2024 season, 98.8% of the reported prices were exactly Rs 2,320 per quintal, the Grade-A MSP for that year. The lowest share in any season was 96.1%, in 2023. Prices also differed very little between mandis on the same day: in 2024 the mean daily standard deviation across mandis was Rs 0.4 per quintal.

With prices this close to a policy value, the MSP rule, which predicts the announced Grade-A MSP for every mandi and date, was the most accurate forecaster we tested. Its mean absolute error over the three test seasons and horizons of 1 to 7 trading dates was Rs 1.26 per quintal. Repeating the last reported price came next at Rs 1.77. The best learned model was A3T-GCN at Rs 2.36, ahead of the GRU without a graph (Rs 2.51) and T-GCN (Rs 2.54). [Add the full error table (Table 3 in v2) generated by script, with seed SDs.]

The date-by-date comparison at a horizon of one trading date shows why. A3T-GCN had a larger error than the naive forecast on 158 of the 158 test dates. Whenever a mandi stays at the MSP, the naive forecast is exactly right, and a neural network almost never predicts a change of exactly zero. The MSP rule tied with the naive forecast on 126 dates, was better on 13 and worse on 19; its misses were smaller, so its mean error was lower. [No significance test was run in v2. Add Diebold–Mariano tests in v3 (R4) before writing "better" as a statistical claim.]

Table 3. Forecast pairs in the held-out season 2025–26 (index with dated rows only): all pairs, and pairs whose origin has at least ten index days of the season.

Variety, horizon	1 day	7 days	14 days	28 days
1121, all pairs	76	73	66	59
1121, eligible	68	64	60	50
1509, all pairs	71	66	60	47
1509, eligible	62	57	51	38

Table 4. Test-season MAE (Rs per quintal) by method, variety and horizon. Filled from the results file only.

Method (1121)	1 day	7 days	14 da
Naive	**??fc_1121_1_naive_mae??**	**??fc_1121_7_naive_mae??**	**??fc_1121_14_naive_mae**
Seasonal naive	**??fc_1121_1_seasonal_naive_mae??**	**??fc_1121_7_seasonal_naive_mae??**	**??fc_1121_14_seasonal_naive_mae**
Five-day average	**??fc_1121_1_ma5_mae??**	**??fc_1121_7_ma5_mae??**	**??fc_1121_14_ma5_mae**

For Paddy (Common), we therefore use the announced MSP as the price in the selling plan and do not train a forecaster. We keep the learned models for basmati paddy, whose prices move with the market (Sect. 4.2).

4.2 Basmati Paddy: the Pre-Registered Forecast Comparison

DRAFT FOR AUTHOR REWRITE. The comparison of Sect. 3.2 has **not been run**. The protocol is a draft until the authors approve it and the hand check has fixed the primary index variant; running any model on these prices before that would break the pre-registration. This subsection therefore reports only what does not depend on forecast error: the data and the number of forecast pairs. The results table is generated from `results/C0_protocol/test_season_results.csv` and stays empty (`??key??`) until that file exists.

The variety index has 433 days for 1121 and 369 days for Basmati 1509 over five seasons, because index days exist only from September to January. This limits what can be tested. Table 3 shows how many forecast pairs the held-out season offers after the rule that an origin needs ten index days of its season. Basmati 1509 has 57 pairs at 7 days and 51 at 14 days, fewer than the 60 first planned, and 38 at 28 days, below the 40 pairs the protocol requires before a coverage claim is made. A power check on the earlier seasons (exploratory, disclosed in the protocol) puts the smallest detectable improvement over the naive forecast at about 11 to 17% of its error. A model that improves on the naive value by less than that cannot be told apart from it with this much data, and "the naive forecast is hard to beat" is a valid and reportable outcome.

4.3 Allocation: Fair Plans Against Simple Rules

Before it was compared with anything, the solver was checked against a brute-force enumeration that shares only the input types and the exact objective with it. On 324

Table 5. Selling policies on 60 synthetic plans at real 2025 prices. Valid: share of plans accepted by the independent validator (a plan is invalid if it exceeds a mandi's daily limit). Net: mean net return in Rs per quintal after transport and waiting costs. Min. share: smallest share of a farmer's lot that was sold, averaged over plans. Fairness price: net return given up relative to the utilitarian plan.

Policy	Valid (%)	Net	Min. share (%)	Gini	Fairness price	Solve (s)
Nearest mandi	100	2,329.69	0.0	0.3125	14.99	0.00
Independent best	0	2,360.49	100.0	0.0000	–	0.00
Utilitarian	100	2,344.68	0.0	0.3087	0.00	0.07
Max-min only	100	2,339.87	68.6	0.0019	4.81	0.57
Greedy	100	2,330.60	68.6	0.0019	14.08	0.07
KisanNiti	100	2,338.98	68.6	0.0019	5.70	4.94
Actual-price bound	100	2,338.98	68.6	0.0019	5.70	5.07

random small instances, three back ends (HiGHS, SCIP and CP-SAT) and two leximin methods were run; 1,728 of 1,728 runs reached exactly the same complete objective as the enumeration, and an instance the enumeration proved infeasible was reported as proven infeasible by every solver.

The allocation method was compared with simple selling rules on 60 weekly plans: ten random draws of a synthetic FPO of 300 members whose lots become ready over six harvest weeks, scored at real 2025 AGMARKNET prices of Paddy (Common) and real mandi coordinates. Every plan was re-checked by the independent validator. The compared rules are the nearest mandi on the first free day, each farmer's independent best choice, a plan that maximises only the total net return (utilitarian), one that maximises only the smallest share, a greedy water-filling heuristic, the KisanNiti hierarchy, and the same hierarchy with the actual prices instead of the forecast (a bound).

Independent choices break a mandi's daily limit in 100% of the plans, so none of them is a valid plan. The nearest-mandi rule respects the limits but leaves the least-served farmer with a share of 0.0%, and the utilitarian plan does the same (0.0%), because maximising the total ignores who is left out. The KisanNiti hierarchy raises the smallest share to 68.6%, the highest that the mandi limits allow, with a Gini coefficient of 0.0019, at a cost of Rs 5.70 per quintal against the utilitarian plan. The greedy rule reaches the same smallest share but gives up Rs 14.08 per quintal, so the exact method recovers the difference. Using actual instead of forecast prices changes nothing, which is expected: the prices of Paddy (Common) hardly differ between mandis (Sect. 4.1), so the plan is decided by distance, limits and fairness, not by price. Of the 240 exact solves, 240 were proven optimal. Three weighted-sum alternatives were also tried in a pilot of six plans; all reached the 300-second time limit without a proof of optimality and are therefore left out of Table 5. The experiment uses a synthetic FPO, so it shows properties of the method and not a gain for any real farmer; the comparison with basmati prices, where mandis differ, is future work. [Add the solver comparison and the sensitivity to transport cost and limits when they are run.]

4.4 Storage Rule on Simulated Scenarios

DRAFT FOR AUTHOR REWRITE. Claude draft, 22 Oct 2026. All readings in this subsection are simulated. Add the real-node results (salt offset, settling time, outage tests, one-week grain run) when the kit is built (`firmware/BUILD_GUIDE.md`).

We have not yet run the node in real grain, so we tested the rule on simulated sensor streams for basmati paddy stored on a farm near Patiala. Air temperature follows the monthly mean daily maximum and minimum of the IMD station at Patiala for 1991–2020[Cite the IMD 1991–2020 climate normals for Patiala: no verified reference on file yet.]. Shed humidity follows a daily cycle fitted through IMD's two monthly means, taken at 08:30 and 17:30 IST; we assume shed air equals outdoor air. The grain's temperature follows the shed air with a 2-day lag, and its water activity drifts slowly towards the shed's daily mean humidity, as it would through a jute bag. These are assumptions, listed in `layers/L3b_storage_iot/scenarios.py`. The 6 scenarios cover grain stored dry and held from November to February, grain held into the monsoon, a lot probed straight after harvest and sun-dried on day 8, the monsoon case with Wi-Fi and power outages, a probe left hanging in shed air, and grain sitting at $a_w \approx 0.65$. Each ran with 20 noise seeds, with no sensor bias and with a bias of ±2 %RH and ±0.3 °C. Patiala's winter mornings are humid (mean RH at 08:30 of 84 % or more from November to January), so in our model even grain stored dry reached $a_w = 0.788$ by February; how fast this happens in a real shed depends on the bag and the shed, which we have not measured.

We compared STORAGE-DEMO-0.1 with three simpler rules: the same thresholds on the latest reading only (no median, no hysteresis), the same thresholds on shed-air humidity, and temperature bands alone. Each rule is scored against the category that the rule's definition gives for the noise-free grain values at the same hour. This measures how much noise, outages, placement and short cuts move a rule away from that definition. It does not show that the definition predicts spoilage.

Table 6 and Fig. 3 give the results. Without sensor bias, STORAGE-DEMO-0.1 agreed with the intended category on 88–100 % of known hours in the four scenarios with the probe in grain and moving conditions, and changed category at most 0.7 times per week. The latest-reading rule agreed about as often (92–99 %) but changed 1.4–19.1 times per week, mostly when the grain sat close to a limit. A shed-air reading was a poor stand-in for the grain: it agreed on 19–61 % of hours and changed 14.4–26.7 times per week, following the daily humidity cycle. Temperature alone marked 100 % of the monsoon hold HIGH and gave no HIGH hour in the winter hold, when moisture rather than heat was the problem (agreement 13 %). When the probe hung in shed air, our rule returned UNKNOWN for every hour, as intended.

The median and the start-up checks have a cost. In the winter, monsoon and outage scenarios the 24-hour median reported HIGH about 12 h after the intended category. A lot probed straight after harvest was marked HIGH 23 h late, because the first day after insertion is UNKNOWN, and after it was dried the rule stayed HIGH for two more days, which made 9 % of its known hours a false HIGH. With the outages, the rule was UNKNOWN for 7 % of the hours, against 4 % for the latest-reading rule; 144 readings were lost, half to the buffer limit during a 30-hour Wi-Fi outage and half to a 6-hour power cut. For grain sitting at $a_w \approx 0.65$, hysteresis held MEDIUM with no changes,

which agreed with the intended category on only 49 % of hours; the latest-reading rule agreed on 92 % but changed 19.1 times per week. Here the rule keeps the cautious level.

A ±2 %RH sensor error mattered about as much as the choice of rule. With +2 %RH, agreement in the four in-grain scenarios fell to 66–96 % and up to 29 % of hours were a false HIGH; with −2 %RH it fell to 60–85 % and up to 15 % of hours were a missed HIGH. The latest-reading rule was affected in the same way, so the median does not protect against bias. We therefore treat the salt calibration (75.3 %RH over saturated NaCl at 25 °C [21]) as a required step before real readings are interpreted.

Table 6. Storage rules on six simulated basmati scenarios (20 seeds, no sensor bias). Agreement is the share of known hours on which a rule gives the intended category; changes are category changes per week; Unknown is the share of hours with no category. Simulated data: rule behaviour only, not spoilage.

	Agreement (%)				Changes per week				Unknown (%)			
Scenario	DEMO	Instant	Shed RH	Temp.	DEMO	Instant	Shed RH	Temp.	DEMO	Instant	Shed RH	Temp.
S1 held Nov–Feb	99	98	55	13	0.1	4.1	26.7	2.5	1	0	0	0
S2 monsoon hold	100	99	58	59	0.1	1.4	16.5	0.0	1	0	0	0
S3 fresh, then dried	88	95	30	52	0.7	11.3	24.7	2.0	3	0	0	0
S4 outages	99	98	61	60	0.2	3.7	14.4	0.0	7	4	4	4
S5 probe in air	–	28	27	30	0.0	26.9	25.2	23.6	100	0	0	0
S6 near 0.65	49	92	19	48	0.0	19.1	25.0	2.5	2	0	0	0

4.5 The Built System

The five stages were built as one product and tested together. The backend, the shared algorithms, the sensor ingestion and the interface contracts hold 1,372 automated tests (backend 319, core 563, ingestion 49, contract examples 441), which include parallel approvals on PostgreSQL, isolation between two FPOs, append-only records and the exact rendering of every message text. The interface is fixed by 12 versioned contracts with 137 valid and 223 invalid examples that every change must still satisfy. A single command runs 14 steps on two new fictional FPOs through the same interface the website uses: data, stock, a simulated storage sensor, plan, edit, approval, messages, the manager's hand-over to WhatsApp, sales, release, a ledger audit and isolation. Each step is checked against an independent recomputation of the amounts, the rounding, the stock ledger and the message texts, and the run stops at the first deviation. A full run takes about 18 seconds and passed all 14 steps. To show that the run can fail, a corrupted ledger entry and deliberately planted errors (rounding down, a wrong weighted price, a message that loses its last character) were each detected. All farmers, lots and sales in these runs are fictional. No user study has been done, no real message was sent, and the Punjabi and Hindi texts have not yet been read by a native speaker.

5 Conclusion and Future Work

DRAFT FOR AUTHOR REWRITE. Drafted by Claude from the results above; the forecasting sentences are completed after the pre-registered run. The authors rewrite this section.

Selling decisions of an FPO depend on whether the crop's price is set by policy or by the market. For Paddy (Common) in Punjab it is set by policy: the announced support price was the most accurate forecast, no learned model beat it, and the useful work is to share the mandi limits fairly. The exact allocation did that, raised the smallest share a farmer receives from 0.0% to 68.6% in the synthetic experiment, and recovered part of what a greedy rule gives up. For basmati the forecast comparison was fixed in advance; its result, including the possibility that nothing beats the naive value, is reported in Sect. 4.2. The five stages from data to recorded sales were built as one system and tested end to end, including an approval that reserves stock exactly and a way for the manager to send each farmer the message from his own WhatsApp.

Limits. The mandi index is a market summary, not the price a farmer received. Index days exist only from September to January, five seasons in all, so the forecast comparison can detect only large improvements. The cleaned prices have not yet passed the hand check. The FPO in the allocation experiment is synthetic, with real prices, so no income gain is claimed. The storage sensor was tested on simulated scenarios, and the node has not run in real grain. Held wheat is not forecast because Punjab mandis report too few wheat prices outside the procurement window. The system sends nothing by itself: the manager's WhatsApp hand-over cannot see delivery, and the Punjabi and Hindi texts have not been checked by a native speaker. Off-plan sales are handled by a provisional rule. No user study has been done.

Future work. A one-week run of the sensor in grain; a hold-or-sell rule with ranges, evaluated per season without a significance claim; the allocation comparison with basmati prices and with real transport costs; a deployment with one FPO after ethics approval, native-speaker review of the messages and a legal review of the privacy controls; and the official messaging channel, which would let one action reach every farmer.

Acknowledgments. [Funding, lab or FPO partners to thank. Write "none" if there are none.]

Use of AI tools. **DRAFT FOR AUTHOR REWRITE.** Wording to be approved by the supervisor before submission.

We used Claude (Anthropic), a large language model, to draft parts of the text from our result files, to write the scripts that generate the figures and tables, and to check that every number in the text matches the result files. The authors rewrote all text in their own words, checked each number and reference against its source, and take full responsibility for the content of this paper.

Data and code availability. [Repository link or archive DOI (e.g. Zenodo) once the repositories are tagged. AGMARKNET data are public; FPO member data are synthetic.]

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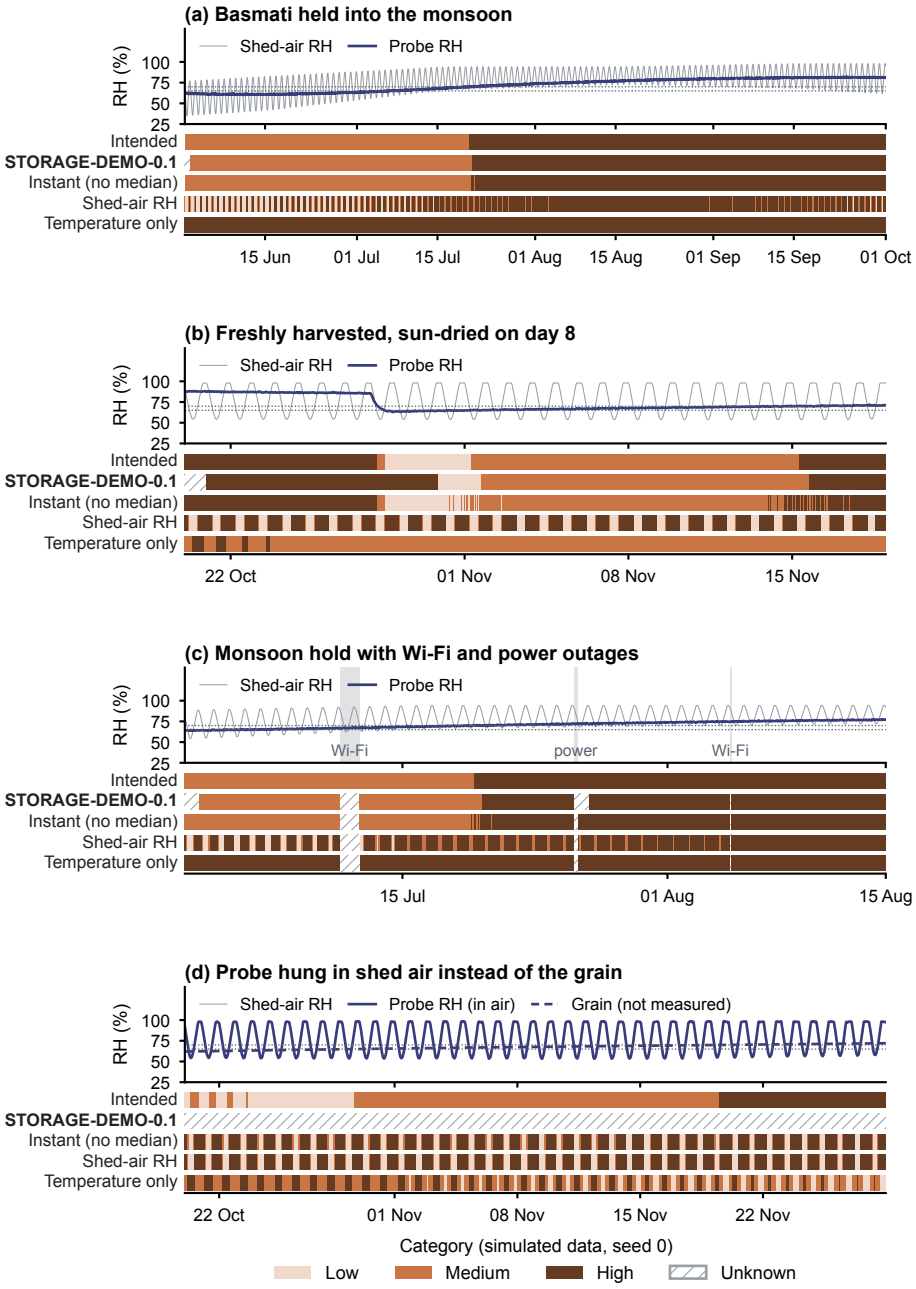


Fig. 3. Simulated storage scenarios (seed 0). Top of each panel: relative humidity read by the probe and of the shed air; dotted lines mark 65 and 70 %. Strips: category per hour for the intended definition, STORAGE-DEMO-0.1 and three simpler rules. (a) Monsoon hold; (b) fresh lot, sun-dried on day 8; (c) monsoon hold with outages (grey bands); (d) probe hung in shed air. Simulated data; the categories describe conditions, not measured spoilage.